Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf____

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ___Serial 0/0/1____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? __Typically **129**

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Does R2 show up as an OSPF neighbor on R1? __No

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Does R2 show up as an OSPF neighbor on R3? __No

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What does this information tell you?

A passive interface does not send or receive OSPF Hello packets, preventing neighbor adjacencies, but its connected network is still advertised.

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

_R2(config)# router ospf 1 R2(config-router)# no passive-interface serial 0/0/1_______

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial 0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65. It is calculated by adding the outgoing Serial interface cost (64) and the destination LAN interface cost (1).

Is R2 listed as an OSPF neighbor to R3? ___Yes___ 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

To differentiate high-speed links (>100Mbps). The default assigns a cost of 1 to all links 100Mbps+, treating them equally. 2

Step 2: Change the bandwidth for an interface.

g.Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

Sum of outgoing interface costs. Formula: 100,000,000 / Bandwidth (bps). 2

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

Cost increases to 782. The command lowers bandwidth to 128Kbps. New cost = 100,000,000 / 128,000 = 781 (plus destination cost). 0

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF selects the lowest cumulative cost. After metric changes, the path via R2 costs less than the direct path. 4

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   Manually setting the router ID ensures network stability. If left unconfigured, OSPF uses the highest active IP address. If that interface fails, the router selects a new ID, causing the OSPF process to restart, dropping

2. Why is the DR/BDR election process not a concern in this lab?

DR/BDR elections are only needed on multi-access broadcast networks (like Ethernet) where multiple routers share a segment. In this lab, routers connect via Serial interfaces, which are Point-to-Point networks. With only two endpoints per link, OSPF naturally knows its neighbor and doesn't need a DR/BDR.

3.Why would you want to set an OSPF interface to passive?

Setting an interface to passive optimizes bandwidth and improves security. It prevents the router from sending OSPF Hello packets on LANs with no other routers (like PC connections). This stops unauthorized devices from sniffing routing data and frees up CPU and link resources.

4.What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

OSPF uses "Areas" to achieve a hierarchical design. By dividing a network into multiple areas connected to a backbone (Area 0), OSPF isolates topology changes. A failure in one area won't trigger recalculations in others, significantly reducing routing table sizes and memory usage.

5.What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6.A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route will be selected. When a router receives routes to the same destination from different protocols, it ignores metrics and compares the Administrative Distance (AD). EIGRP has an AD of 90, making it more trustworthy than OSPF (AD 110) or RIP (AD 120).